

Computing limits of functions



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- **Book:** “Analisi matematica 1”, Marco Bramanti, Carlo D. Pagani and Sandro Salsa, Zanichelli.

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1 Computing limits of functions

- We state the theorems on limits of functions that follow immediately from the corresponding theorems on limits of sequences and from the sequential definition of limit

1.1 Algebra of limits theorem

Theorem 1: Algebra of limits, case of finite limits

Hypotheses as $x \rightarrow c$:

$$1. f(x) \rightarrow \ell_1 \in \mathbb{R}, \quad 2. g(x) \rightarrow \ell_2 \in \mathbb{R}.$$

Thesis as $x \rightarrow c$:

$$1. f(x) \pm g(x) \rightarrow \ell_1 \pm \ell_2, \quad 2. f(x) g(x) \rightarrow \ell_1 \ell_2,$$

$$3. \frac{f(x)}{g(x)} \rightarrow \frac{\ell_1}{\ell_2} \quad (\ell_2 \neq 0, g(x) \neq 0, \text{ eventually as } x \rightarrow c).$$

Proof. Let $\{x_n\}$ be any sequence such that

$$x_n \neq c, \forall n, \text{ and } x_n \rightarrow c \text{ as } n \rightarrow +\infty.$$

By hypothesis we have

$$f(x_n) \rightarrow \ell_1 \text{ and } g(x_n) \rightarrow \ell_2 \quad (\ell_1, \ell_2 \in \mathbb{R}).$$

From the theorem on the algebra of limits for sequences we therefore conclude that

$$f(x_n) \pm g(x_n) \rightarrow \ell_1 \pm \ell_2,$$

and hence

$$f(x) \pm g(x) \rightarrow \ell_1 \pm \ell_2.$$

Claims 2 and 3 of the theorem are proved in a perfectly analogous way.

□

1.2 Sign-preservation theorems

In the following statements ℓ and c will be points of $\mathbb{R}^*$ unless otherwise stated.

Theorem 2: Sign preservation, 1st form

Hypotheses:

$$1. f(x) \rightarrow \ell \text{ as } x \rightarrow c, \quad 2. \ell \leq 0.$$

Thesis:

$$f(x) \leq 0 \text{ eventually as } x \rightarrow c.$$

Proof. Let $\{x_n\}$ be any sequence such that

$$x_n \neq c, \forall n, \text{ and } x_n \rightarrow c \text{ as } n \rightarrow +\infty$$

By the hypothesis we have

$$f(x_n) \rightarrow \ell > 0$$

hence, by the sign-preservation theorem for sequences, applied to the sequence $\{f(x_n)\}$, we conclude that

$$f(x_n) > 0, \text{ eventually.}$$

Since this holds for every sequence such that $x_n \rightarrow c$, we conclude that

$$f(x) > 0, \text{ eventually, as } x \rightarrow c$$

□

Theorem 3: Sign preservation for functions, 2nd form

Hypotheses:

$$1. f(x) \rightarrow \ell \in \mathbb{R} \text{ as } x \rightarrow c, \\ 2. f(x) \geq 0 \text{ eventually as } x \rightarrow c.$$

Thesis:

$$\ell \geq 0.$$

Proof. It follows from the corresponding theorem for sequences.

□

- For functions we also have the following sign-preservation theorem.

Theorem 4: Sign preservation for continuous functions

Hypotheses:

1. f is continuous at $c \in \mathbb{R}$,
2. $f(c) > 0$.

Thesis:

$$f(x) > 0 \text{ eventually as } x \rightarrow c.$$

Proof. If f is continuous at c , then

$$f(c) = \lim_{x \rightarrow c} f(x)$$

hence the hypothesis $f(c) > 0$ means that

$$f(c) = \lim_{x \rightarrow c} f(x) > 0$$

and this, by the sign-preservation theorem (1st form), implies that

$$f(x) > 0 \text{ eventually as } x \rightarrow c.$$

□

1.3 Comparison theorem

Theorem 5: Comparison theorem

Hypotheses:

1. $f(x) \rightarrow \ell$ and $g(x) \rightarrow \ell$ as $x \rightarrow c$,
2. $f(x) \leq h(x) \leq g(x)$ eventually as $x \rightarrow c$.

Thesis:

$$h(x) \rightarrow \ell \text{ as } x \rightarrow c.$$

Proof. Let $\{x_n\}$ be any sequence such that

$$x_n \neq c, \forall n, \text{ and } x_n \rightarrow c \text{ as } n \rightarrow +\infty$$

We want to prove that

$$h(x_n) \rightarrow \ell \text{ as } n \rightarrow +\infty$$

By hypothesis we know that:

$$f(x_n) \leq h(x_n) \leq g(x_n), \text{ eventually,}$$

$$f(x_n) \rightarrow \ell \text{ and } g(x_n) \rightarrow \ell, \text{ as } x_n \rightarrow c$$

Hence, by the comparison theorem for sequences applied to:

$$\{f(x_n)\}, \{h(x_n)\} \text{ and } \{g(x_n)\}$$

we conclude that

$$h(x_n) \rightarrow \ell \text{ as } n \rightarrow +\infty$$

□

Corollary 1: Of the comparison theorem (part I)

Hypotheses:

1. $g(x) \rightarrow 0$ as $x \rightarrow c$,
2. $|h(x)| \leq g(x)$ eventually as $x \rightarrow c$.

Thesis:

$$h(x) \rightarrow 0 \text{ as } x \rightarrow c.$$

Proof. It follows from the corresponding corollary for sequences.

□

Example 1: Corollary of the comparison theorem

Let us prove that:

$$\lim_{x \rightarrow 0} x \sin \frac{1}{x} = 0$$

We have

$$x \rightarrow 0 \text{ as } x \rightarrow 0 \text{ and } \lim_{x \rightarrow 0} \sin \frac{1}{x} \text{ does not exist}$$

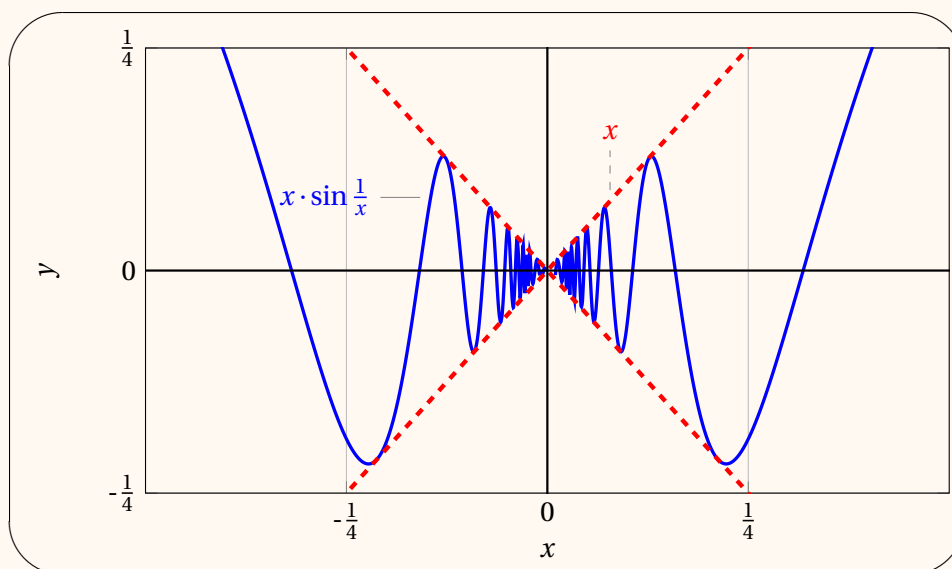
hence we cannot apply Theorem (??) on the algebra of limits for functions.

We have

$$\left| \sin \frac{1}{x} \right| \leq 1 \text{ hence } \left| x \sin \frac{1}{x} \right| \leq |x| \text{ and } |x| \rightarrow 0 \text{ as } x \rightarrow 0.$$

Hence, by Corollary (??) of the comparison theorem for functions, we have:

$$x \sin \frac{1}{x} \rightarrow 0 \text{ as } x \rightarrow 0.$$



Corollary 2: Of the comparison theorem (part II)

Hypotheses:

1. $f(x) \rightarrow 0$ as $x \rightarrow c$,
2. $g(x)$ is bounded, eventually, as $x \rightarrow c$.

Thesis:

$$f(x) g(x) \rightarrow 0 \text{ as } x \rightarrow c.$$

Proof. It follows from the corresponding corollary for sequences. □

Example 2: Corollary of the comparison theorem

Let us prove that:

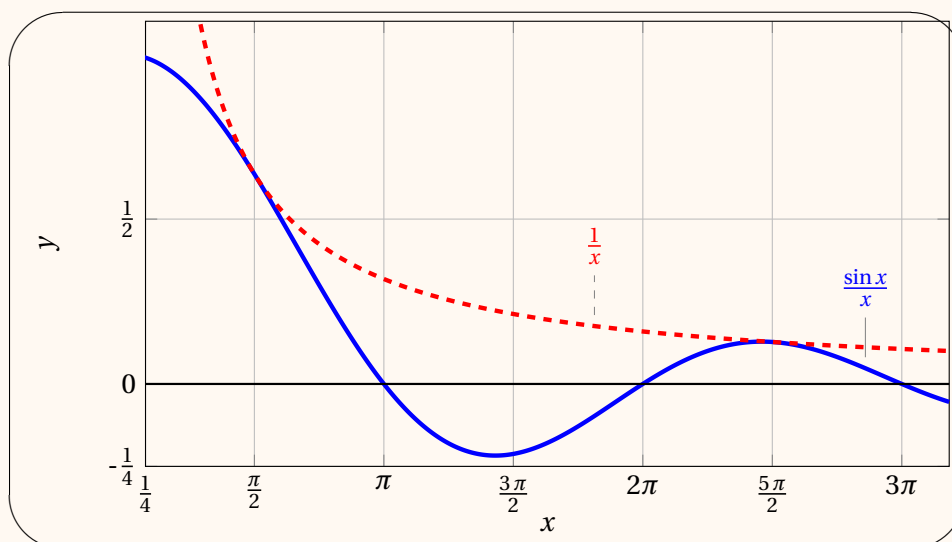
$$\lim_{x \rightarrow +\infty} \frac{x + \sin x}{2x + \cos x} = \frac{1}{2}$$

Factoring, we obtain

$$\frac{x + \sin x}{2x + \cos x} = \frac{x \left(1 + \frac{\sin x}{x}\right)}{2x \left(1 + \frac{\cos x}{2x}\right)} = \frac{1}{2} \left(\frac{1 + \frac{\sin x}{x}}{1 + \frac{\cos x}{2x}} \right)$$

By Corollary (??) of the comparison theorem for functions, we have:

$$\frac{\sin x}{x} \rightarrow 0 \text{ and } \frac{\cos x}{2x} \rightarrow 0 \text{ as } x \rightarrow +\infty.$$



Then, by Theorem (??) on the algebra of limits, we therefore have:

$$\frac{x + \sin x}{2x + \cos x} \rightarrow \frac{1}{2} \text{ as } x \rightarrow +\infty.$$

1.4 Theorems of partial arithmetization of the infinity symbol

Theorem 6: Partial arithmetization of the infinity symbol (addition)

Hypotheses as $x \rightarrow c$:

$$1. f(x) \rightarrow \ell \in \mathbb{R}, \quad 2. g(x) \rightarrow +\infty, \quad 3. h(x) \rightarrow +\infty.$$

Thesis as $x \rightarrow c$:

$$\begin{aligned} 1. f(x) + g(x) &\rightarrow \ell + \infty = +\infty, & 2. f(x) - g(x) &\rightarrow \ell - \infty = -\infty, \\ 3. g(x) + h(x) &\rightarrow +\infty + \infty = +\infty, & 4. -g(x) - h(x) &\rightarrow -\infty - \infty = -\infty. \end{aligned}$$

Proof. It follows from the corresponding theorem for sequences. □

Theorem 7: Partial arithmetization of the infinity symbol (product)

Hypotheses as $x \rightarrow c$:

$$1. f(x) \rightarrow \ell \in \mathbb{R}, \quad 2. g(x) \rightarrow 0, \quad 3. h(x) \rightarrow \infty.$$

Thesis as $x \rightarrow c$:

$$\begin{aligned} 1. f(x) h(x) &\rightarrow \ell \cdot \infty = \infty \quad (\ell \neq 0), & 2. \frac{f(x)}{g(x)} &\rightarrow \frac{\ell}{0} = \infty \quad (\ell \neq 0), \\ 3. \frac{f(x)}{h(x)} &\rightarrow \frac{\ell}{\infty} = 0. \end{aligned}$$

Proof. It follows from the corresponding theorem for sequences. □

- As for sequences, the **sign** of ∞ must be determined with the **rule of signs**.

Example 3: Partial arithmetization of the infinity symbol

$$\lim_{x \rightarrow -\infty} \left(\frac{1}{x} - 2 \right) x^3 = +\infty$$

Since $\left(\frac{1}{x} - 2 \right) \rightarrow -2$ and $x^3 \rightarrow -\infty$ as $x \rightarrow -\infty$.

2 Change of variable theorem for limits

Theorem 8: Change of variable in the limit

Let f and g be two functions for which the composition $f \circ g$ is defined, at least eventually as $x \rightarrow x_0 \in \mathbb{R}^*$. If:

1. $\lim_{x \rightarrow x_0} g(x) = t_0 \in \mathbb{R}^*$, 2. $\lim_{t \rightarrow t_0} f(t) = \ell \in \mathbb{R}^*$
3. $g(x) \neq t_0$, eventually, as $x \rightarrow x_0$

Then:

$$\lim_{x \rightarrow x_0} f(g(x)) = \lim_{t \rightarrow t_0} f(t).$$

Hypothesis 3. is not necessary when f is continuous at t_0 , or when $t_0 = \pm\infty$.

Proof. Let $\{x_n\}$ be any sequence such that

$$x_n \neq x_0, \forall n, \text{ and } x_n \rightarrow x_0 \text{ as } n \rightarrow +\infty.$$

We have

$$g(x_n) \rightarrow t_0 \text{ as } n \rightarrow +\infty \text{ (by hypothesis 1)}$$

and

$$g(x_n) \neq t_0, \text{ eventually (by hypothesis 3).}$$

Therefore

$$f(g(x_n)) \rightarrow \ell \text{ (by hypothesis 2).}$$

If $t_0 = \pm\infty$ the condition $g(x) \neq \pm\infty$ is obviously satisfied, while if f is continuous at t_0 , $\ell = f(t_0)$, so if $g(x_n) = t_0$ for some n we would have

$$f(g(x_n)) = f(t_0) = \ell$$

and hence the convergence

$$f(g(x_n)) \rightarrow \ell \text{ as } n \rightarrow +\infty$$

would be guaranteed anyway.

□

Example 4: Computing a limit with the change of variable theorem

Let us compute the limit

$$\lim_{x \rightarrow +\infty} \log \left(\frac{2x^3 + 4x + 1}{5(x+1)^3} \right)$$

The functions f and g are:

$$g(x) = \left(\frac{2x^3 + 4x + 1}{5(x+1)^3} \right) \quad \text{and} \quad f(t) = \log t.$$

We have $t = g(x)$ and

$$(f \circ g)(x) = f(g(x)) = \log \left(\frac{2x^3 + 4x + 1}{5(x+1)^3} \right) \quad \text{and} \quad x_0 = +\infty.$$

We compute

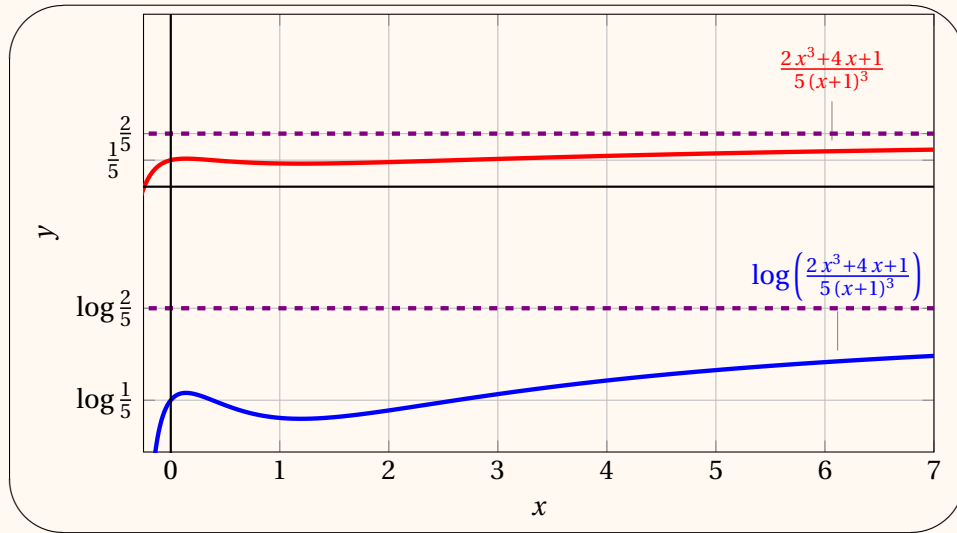
$$\lim_{x \rightarrow +\infty} g(x) = \frac{2}{5} \quad \text{and consequently} \quad t_0 = \frac{2}{5}.$$

Now we compute

$$\lim_{t \rightarrow 2/5} \log t = \log \frac{2}{5} \quad (\text{as we will see, the logarithm is continuous in } \mathbb{R}_+)$$

Hence

$$\lim_{x \rightarrow +\infty} f(g(x)) = \log \frac{2}{5}$$



Moreover:

$$\lim_{x \rightarrow 0} f(g(x)) = \log \frac{1}{5}$$

using the theorem we have:

$$\lim_{x \rightarrow 0} g(x) = \frac{1}{5} \quad \text{and} \quad \lim_{t \rightarrow 1/5} \log t = \log \frac{1}{5}$$

since

$$\lim_{x \rightarrow 0} 2x^3 + 4x + 1 = 1 \quad \text{and} \quad \lim_{x \rightarrow 0} 5(x+1)^3 = 5$$

Example 5: Computing a limit with the change of variable theorem

Let us compute the limits

$$\lim_{x \rightarrow 0^+} e^{\frac{1}{x}} \quad \text{and} \quad \lim_{x \rightarrow 0^-} e^{\frac{1}{x}}.$$

The functions f and g are:

$$g(x) = \frac{1}{x} \quad \text{and} \quad f(t) = e^t.$$

We have $t = g(x)$ and

$$(f \circ g)(x) = f(g(x)) = e^{\frac{1}{x}} \quad \text{moreover} \quad x_0 = 0^+ \quad \text{and} \quad x_0 = 0^-.$$

1. $\frac{1}{x} \rightarrow +\infty$ as $x \rightarrow 0^+$ and hence $t_0 = +\infty$. Now we compute:

$$\lim_{t \rightarrow +\infty} e^t = +\infty \quad \text{hence} \quad \lim_{x \rightarrow 0^+} e^{\frac{1}{x}} = +\infty.$$

2. $\frac{1}{x} \rightarrow -\infty$ as $x \rightarrow 0^-$ and hence $t_0 = -\infty$. Now we compute:

$$\lim_{t \rightarrow -\infty} e^t = 0 \quad \text{hence} \quad \lim_{x \rightarrow 0^-} e^{\frac{1}{x}} = 0.$$

